Carr-Madan Spanning Formula & VIX

Math 490 - Research Project

Artem Ivaniuk, Andy Chang, Krishaan Chaudhary

May 6, 2026

Carr-Madan: Idea

Goal: replicate nonlinear option payoff with a portfolio of options

$$g(S_t) =$$

$$g(F) + g'(F)(S_t - F) + \int_0^F g''(K)(K - S_t)^+ dK + \int_F^\infty g''(K)(S_t - K)^+ dK$$

- $g(S_t)$ = value of derivative
- F is a constant
- European style payoff - payoff at expiration
- Enables a static hedge of complex options using puts and calls

Carr-Madan: Derivation Intuition

$$g(x) = g(a) + \int_a^x g'(t)dt$$

$$g(x) = g(a) + \int_a^x [g'(t) + g'(a) - g'(a)]dt$$

$$g(x) = g(a) + g'(a)(x-a) + \int_a^x [g'(t) - g'(a)]dt$$

$$g(x) = g(a) + g'(a)(x-a) + \int_a^x \left[\int_{u=a}^{u=t} g''(u)du\right] dt$$

$$g(x) = g(a) + g'(a)(x-a) + \int_a^x \left[\int_{t=u}^{t=x} g''(t)dt\right] du$$

$$g(x) = g(a) + g'(a)(x-a) + \int_a^x g''(u)(x-u)du$$

- FTC, Adding Zero, Integrate constant, Rewrite as Double Integral, Fubini's Theorem

Carr-Madan: Derivation Intuition

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases} \qquad H(x) + H(-x) = 1$$

$$\int_a^x g''(u)(x-u)du = \int_a^x [H(x-a) + H(a-x)]g''(u)(x-u)du$$

$$= \int_a^x H(x-a)g''(u)(x-u)du + \int_x^a H(a-x)g''(u)(u-x)du$$

$$\cdots = \int_a^\infty (x-u)^+ g''(u)du + \int_{-\infty}^a (u-x)^+ g''(u)du$$

- Heaviside Function, Symmetry, Extending Bounds, Piecewise Function Properties

VIX as a Target

VIX measures:

$$\frac{1}{T}\mathbb{E}\left[\int_0^T \sigma_t^2 dt\right]$$

Interpretation:

- Risk-neutral expected variance
- Derived from S&P500 options across past 30 days' strikes
- Option prices encode volatility
- Described as a "fear gauge"
- Although not directly traded, there are futures and ETFs
- Can develop a replicating portfolio to hedge against uncertainty
- Can be used in trading strategies for insurance against downturn or a way to trade volatility itself

VIX During 2008 Financial Crisis

VIX Formula

$$\text{VIX}^2 \approx \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i)$$

- $Q(K_i)$: OTM (Out of the Money) option prices
- OTM options encapsulate market sentiment about extreme changes
- puts for $K_i < F$, calls for $K_i > F$
- ΔK_i: strike spacing
- $\frac{2}{T}$: scaling factor for time
- $\frac{1}{K_i^2}$: lower strike prices weighted more

Log Dynamics → Expected Variance

$$d\log S_t = \left(r - \frac{1}{2}\sigma_t^2\right)dt + \sigma_t dW_t$$

$$\log S_T - \log S_0 = \int_0^T \left(r - \frac{1}{2}\sigma_t^2\right)dt + \int_0^T \sigma_t dW_t$$

$$\mathbb{E}[\log S_T] = \log S_0 + rT - \frac{1}{2}\mathbb{E}\left[\int_0^T \sigma_t^2 dt\right]$$

$$\mathbb{E}\left[\int_0^T \sigma_t^2 dt\right] = -2\,\mathbb{E}\left[\log\left(\frac{S_T}{F}\right)\right]$$

- Ito's Lemma converts price dynamics into log form with variance appearing explicitly → we now can isolate variance
- Stochastic integral drops out under expectation (martingale)
- Isolating integrated variance as a function of log-returns
- Forward price $F = S_0 e^{rT}$ simplifies expectation

Eliminating Linear Terms → Pricing Identity

$$f(S_T) = -2\log\left(\frac{S_T}{F}\right), \quad f''(K) = \frac{2}{K^2}$$

$$f(F) + f'(F)(S_T - F) = -\frac{2}{F}(S_T - F)$$

$$\mathbb{E}[S_T] = F \Rightarrow \text{term vanishes}$$

$$\mathbb{E}[f(S_T)] = \int_0^F \frac{2}{K^2} P(K)\, dK + \int_F^\infty \frac{2}{K^2} C(K)\, dK$$

$$\mathbb{E}[f(S_T)] = \mathbb{E}\left[\int_0^T \sigma_t^2 dt\right]$$

- Linear terms cancel under risk-neutral expectation
- Remaining terms are purely option payoffs
- Put and call prices replace expectations of $(\cdot)^+$ payoffs
- We got a link between option prices and expected variance

From Integral to VIX Formula: Intuition

$$\mathbb{E}\left[\int_0^T \sigma_t^2 dt\right] = \int_0^\infty \frac{2}{K^2} Q(K)\, dK$$

$$\int_0^\infty \frac{2}{K^2} Q(K)\, dK \approx \sum_i \frac{2}{K_i^2} Q(K_i)\, \Delta K_i$$

$$\mathrm{VIX}^2 \approx \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i)$$

- Combine puts and calls into unified OTM price function $Q(K)$
- Extend integration over all strikes using payoff symmetry
- Replace integral with discrete sum over traded strikes
- Normalize by maturity T to obtain implied variance (VIX)

Numerical Approximation

For the numerical approximation, we used **yfinance**, a Python api for Yahoo Finance ticker information.
Key points about our approach:

- We used S&P500 index put / call prices to price VIX
- SPX index offers European Options, one of the assumptions needed for the Carr-Madan spanning formula
- We consider the midpoint of the bid / ask prices on an option to be its price.

Numerical Approximation Inputs / Output

Inputs:

- Current Date
- S&P500 Current Price

Run Conditions:

- $r = 4.31\%$ (annual interest rate)
- Timestamp of run: 04-25-2026 9:58pm

Output is the approximated VIX price

Results + Comparison

Figure: Vix price at close on 04-24-26

Code output results:

```
Option Expiry Date:  2026-05-26
S0:  7165.08
Sigma squared term:  0.03508
VIX via Carr-Madan:  18.7309
```

Ta-Dah!

Thank you!
Any questions?